

Syllabus for

Academic year 2020/2021 >

FFM516 - Mechanics 1

Mekanik 1

Syllabus adopted 2019-02-14 by Head of Programme (or corresponding)

Owner: [TKTFY](#)**7,5 Credits****Grading:** TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail**Education cycle:** First-cycle**Major subject:** Engineering Physics**Department:** 16 - PHYSICS**Teaching language:** Swedish**Application code:** 57131**Open for exchange students:** No

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0113 Examination 3,0 c Grading: TH		3,0 c					Tue 12/01-2021 pm J  Wed 07/04-2021 am J, Wed 18/08-2021 am J
0213 Examination 4,5 c Grading: TH			4,5 c				Thu 18/03-2021 am J, Tue 05/01-2021 pm J, Sat 28/08-2021 am J 

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 1 \(compulsory\)](#)[TKTFY ENGINEERING PHYSICS, Year 1 \(compulsory\)](#)

Examiner:

[Ulf Gran](#)[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Courses of the program during the first study period.

Aim

- To provide a good understanding of the basic concepts of mechanics, that provides a basis for further physics studies.
- To provide training in the translation of a physics problem into a mathematical model, which is analysed by application of concepts introduced in the mathematic courses.

Learning outcomes (after completion of the course the student should be able to)

Introduction - The student should understand the central role of mechanics within the natural sciences.

Force systems - The student should be able to apply linear algebra and trigonometry in order to efficiently and reliably analyse (rigid body equivalent) systems of forces and torques.

Equilibrium - The student should be able to divide a system into subsystems, and model interactions between these through forces and torques. The student should also be able to apply this to analyse situations with statically determined equilibrium.

Friction - The student should understand the special properties of the force of friction and be able to apply the Coulomb

model for dry friction in cases of equilibrium, limiting cases of equilibrium, and uniform motion.

Force distributions - The student should have acquired a general understanding of when an interaction needs to be modelled by a force distribution and be able to solve problems related to applications in hydrostatics and thin beams.

Particle kinematics - The student should be able to make use of the most common coordinate systems in 2D and 3D when solving problems.

Particle dynamics - The student should be able to reliably analyse systems of particles in motion, in particular systems with constraints.

Content

Introduction
Force systems
Equilibrium
Friction
Distributed forces
Kinematics of particles
Particle dynamics
Planetary motion

Organisation

Video lectures
Lectures
Problem sessions
Assignments for homework

Literature

The course textbook will be announced on the homepage prior to course start.

Examination including compulsory elements

Written tests in sp 2 and sp 3, consisting of a set of problems to solve, and obligatory hand-ins (almost) every week.